

Software Requirements Specification

Functional and non-functional requirements for the CareBridge platform, each traced to its implementation status

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Table of Contents

1 Introduction	3
1.1 Purpose	3
1.2 Scope	3
1.3 Definitions	3
1.4 Requirement Identifier Scheme	4
2 Overall Description	5
2.1 Product Perspective	5
2.2 User Classes	5
2.3 Operating Environment	5
2.4 Design and Implementation Constraints	5
2.5 Assumptions and Dependencies	6
3 Functional Requirements — Identity and Access	7
4 Functional Requirements — Referral Creation and Matching	9
5 Functional Requirements — Referral Lifecycle	11
6 Functional Requirements — Admissions and Hospital Operations	13
7 Functional Requirements — Laboratory Module	14
8 Functional Requirements — Commission and Settlement	15
9 Functional Requirements — Platform Administration	17
10 Functional Requirements — Notifications, Documents, and Payments	18
11 Non-Functional Requirements	20
11.1 Security	20
11.2 Performance and Scalability	21
11.3 Reliability and Integrity	21
11.4 Usability and Accessibility	22
11.5 Maintainability	22
12 Requirements Summary	24

1. Introduction

1.1 Purpose

This specification records the requirements that the CareBridge platform was built to satisfy, and states the implementation status of each. It serves two purposes at handover: it is the reference against which the delivered system can be verified, and it is the baseline from which any future phase of work is specified.

Every requirement carries a unique identifier and a status. Statuses are used strictly: Implemented means present and operational in the delivered system; Partial means present but dependent on a client action or configuration to become fully operational; Not implemented means absent from the delivered system.

1.2 Scope

CareBridge is a web-based platform that manages the referral of patients from independent consultants to hospitals and diagnostic laboratories, and the commercial settlement that follows treatment. It covers the full path from a consultant creating a referral, through hospital acceptance, admission, treatment, and billing, to the division of the resulting revenue between the facility, the consultant, and the platform operator.

The system does not provide clinical records management, does not prescribe or record treatment beyond free-text clinical notes, and is not a hospital information system. It records the referral relationship and the money that flows from it.

1.3 Definitions

Term	Definition
Consultant	A referring doctor registered on the platform, who creates referrals and earns commission on them.
Facility	A hospital or laboratory that receives referrals.
Referral	A structured request from a consultant to a facility to receive a named patient, carrying clinical and demographic detail.
Admission	The record created when a hospital admits a referred patient, and against which the case is eventually billed.
Response deadline	The time by which a facility must accept or reject a referral, determined by the referral's clinical urgency.
Escalation	Automatic reassignment of a referral to another facility when the response deadline passes without a decision.
Platform charge	The amount the platform operator retains from a billed case.

Term	Definition
Consultant commission	The amount payable to the referring consultant from a billed case.
Settlement	A periodic reconciliation in which a facility pays the platform for a set of billed cases and consultant payouts are released.
Paisa	The subunit of the Pakistani Rupee. One rupee equals 100 paisa. All monetary values are stored as whole paisa.

1.4 Requirement Identifier Scheme

Prefix	Area
FR-AUTH	Identity, registration, and access control
FR-REF	Referral creation and matching
FR-LIFE	Referral lifecycle and escalation
FR-ADM	Admissions and hospital operations
FR-LAB	Laboratory module
FR-FIN	Commission and settlement
FR-PLAT	Platform administration
FR-NOTE	Notifications and real-time updates
FR-DOC	Documents, exports, and payments
NFR	Non-functional requirements

2. Overall Description

2.1 Product Perspective

CareBridge is a self-contained web application with a browser-based client and a server-side API. It depends on four external services: a document database, a file storage service, a transactional email service, and — for optional features — a messaging service and a payment gateway. It does not integrate with hospital information systems, laboratory information systems, or national health registries; verification of professional registration numbers is a manual administrative step supported by document upload.

2.2 User Classes

User class	Characteristics	Technical proficiency
Consultant	Independent referring doctors. Use the system in short bursts between patients, frequently on a mobile device.	Low to moderate. The referral flow is guided step by step for this reason.
Hospital staff	Administrative and reception staff working the referral inbox continuously during shifts, plus managers using billing and settlement screens.	Moderate. Multiple staff share one hospital account tree via sub-accounts.
Laboratory staff	Reception and reporting staff processing test referrals and uploading results.	Moderate.
Platform administrator	CareBridge operations staff performing approvals, configuration, and financial verification.	High. This class has unrestricted access and configures behaviour affecting all other users.

2.3 Operating Environment

- **Client** — any current desktop or mobile browser (Chrome, Edge, Firefox, Safari). No installation is required. The interface adapts to small screens.
- **Server** — Node.js 20 or later, running on a managed hosting platform.
- **Database** — MongoDB, with geospatial indexing enabled for distance-based matching.
- **Network** — the client requires an internet connection; real-time updates require an outbound WebSocket connection to be permitted by the user's network.

2.4 Design and Implementation Constraints

- **Monetary precision** — all money is stored as whole paisa in integer fields. No monetary value may be held as a floating-point number.
- **Single commission engine** — every calculation that splits a bill must use the shared commission module. The design record identifies duplicated bill-splitting arithmetic as the principal historical defect risk in this system.

- **Historical reproducibility** — the commercial terms applied to a case must be recorded on the resulting payout record, so that figures remain reproducible after terms change.
- **Backwards compatibility of commercial terms** — the additive commission model was introduced after the platform was in use. Existing parties default to the original nested model and are opted in individually.
- **Locale** — telephone numbers and identity numbers are validated against Pakistani formats. Times shown in messages are rendered in Pakistan Standard Time. Currency is the Pakistani Rupee.

2.5 Assumptions and Dependencies

- Professional registration numbers are verified by an administrator against uploaded certificates. The system does not verify them against an external registry.
- Facilities maintain their own bed inventory and test catalogue accurately. The platform cannot detect a stale bed count, and matching quality depends directly on this.
- Settlement is receipt-based. Payment between the parties occurs outside the platform and is evidenced by uploaded receipts verified by an administrator.
- Email delivery depends on the third-party provider. Registration cannot complete if verification email delivery fails.

3. Functional Requirements — Identity and Access

ID	Requirement	Status
FR-AUTH-01	The system shall allow consultants, hospitals, and laboratories to register themselves through role-specific registration forms. Administrator accounts shall not be creatable through public registration.	Implemented
FR-AUTH-02	Consultant registration shall capture full name, email, mobile number, password, professional registration number, national identity number, specialty, clinic name and address, and clinic geographic location.	Implemented
FR-AUTH-03	Hospital registration shall capture hospital name, contact details, password, registration number, representative identity number, address, geographic location, operating departments, and initial bed inventory by ward type.	Implemented
FR-AUTH-04	Laboratory registration shall capture laboratory name, contact details, password, registration number, representative identity number, city, area, address, geographic location, and an initial test catalogue with prices and turnaround times.	Implemented
FR-AUTH-05	The system shall require applicants to upload verification documents — a professional or organisational registration certificate and identity documents — before an application can be submitted.	Implemented
FR-AUTH-06	The system shall require the applicant to view and accept the standard operating procedure and privacy policy before submission.	Implemented
FR-AUTH-07	The system shall verify the applicant's email address by a six-digit one-time code, and shall permit the code to be resent subject to a cooldown period.	Implemented
FR-AUTH-08	The system shall create every new account in a pending state and shall prevent sign-in until an administrator activates it.	Implemented
FR-AUTH-09	The system shall allow an administrator to approve an application, or reject it with a reason that is communicated to the applicant.	Implemented
FR-AUTH-10	The system shall notify all administrators when a new registration is submitted.	Implemented
FR-AUTH-11	The system shall authenticate users by email and password, and shall reject sign-in for accounts that are pending, suspended, or whose email is unverified, stating which condition applies.	Implemented
FR-AUTH-12	The system shall store passwords only as salted cryptographic hashes and shall require a minimum length of eight characters.	Implemented
FR-AUTH-13	The system shall issue a short-lived access credential and a longer-lived renewal credential, and shall renew the session transparently so that a user working continuously is not interrupted.	Implemented
FR-AUTH-14	The system shall refuse to renew a session for an account that is no longer active.	Implemented

ID	Requirement	Status
FR-AUTH-15	The system shall allow a user to request a password reset by email, using a single-use link that expires after a limited period.	Implemented
FR-AUTH-16	The system shall not disclose whether an email address is registered when a password reset is requested.	Implemented
FR-AUTH-17	The system shall allow an administrator to set a new password for any user.	Implemented
FR-AUTH-18	The system shall notify a user when their password is changed and when their account is signed into.	Implemented
FR-AUTH-19	The system shall restrict every screen and every endpoint by role, and shall redirect a user attempting to reach a screen outside their role.	Implemented
FR-AUTH-20	The system shall allow hospitals, laboratories, and the platform to create team member accounts that sign in to the same portal as their parent organisation.	Implemented
FR-AUTH-21	The system shall prevent a team member from removing an account that created their own, directly or indirectly.	Implemented
FR-AUTH-22	The system shall allow an administrator to suspend and reactivate an account without deleting its history.	Implemented
FR-AUTH-23	The system shall allow a user to update their own name, mobile number, password, and verification documents, and shall prevent replacement of documents that an administrator has verified.	Implemented

4. Functional Requirements — Referral Creation and Matching

ID	Requirement	Status
FR-REF-01	The system shall provide consultants with a guided multi-step referral intake covering patient details, clinical assessment, location preferences, and facility selection, allowing free movement between steps without loss of entered data.	Implemented
FR-REF-02	The system shall capture patient name, national identity number, guardian name and relation, date of birth or age, gender, and contact number.	Implemented
FR-REF-03	The system shall accept patient age expressed in years, months, or days and shall derive a date of birth from it, so that infants and neonates can be recorded accurately.	Implemented
FR-REF-04	The system shall encrypt the patient national identity number before storage and shall decrypt it only when serving an authorised request.	Implemented
FR-REF-05	The system shall capture presenting symptoms as free text, a clinical summary, and optional file attachments.	Implemented
FR-REF-06	The system shall suggest a clinical department from the entered symptoms, using a keyword catalogue maintained by administrators.	Implemented
FR-REF-07	The system shall require a clinical urgency of emergency, urgent, or routine on every referral.	Implemented
FR-REF-08	The system shall capture the ward type required by the patient.	Implemented
FR-REF-09	The system shall rank candidate hospitals for a referral by a weighted score across department match, bed availability, distance from the patient, cost fit, response-time history, and consultant preference.	Implemented
FR-REF-10	The system shall exclude from the ranking any hospital that does not operate the required department, or that has no available bed in the required ward type.	Implemented
FR-REF-11	The system shall treat intensive care as the required ward for emergency referrals and a general ward otherwise, when applying the bed availability filter.	Implemented
FR-REF-12	The system shall calculate distance between the patient and each hospital using geographic coordinates, and shall award no distance score beyond a 30 kilometre horizon.	Implemented
FR-REF-13	The system shall present, for each recommended hospital, the reasons for its ranking.	Implemented
FR-REF-14	The system shall allow the consultant to select a specific department and a named doctor at the receiving hospital.	Implemented
FR-REF-15	The system shall allow a consultant to submit a referral to more than one hospital, each tracked independently.	Implemented

ID	Requirement	Status
FR-REF-16	The system shall allow a consultant to mark hospitals as favourites, and shall weight favourited hospitals more highly in that consultant's future recommendations.	Implemented
FR-REF-17	The system shall record the number of referrals a consultant has previously sent to each hospital and shall use this history to weight future recommendations.	Implemented
FR-REF-18	The system shall allocate a unique sequential referral code to every referral, in a form that includes the year, and shall allocate codes atomically so that no two referrals can share one.	Implemented
FR-REF-19	The system shall record a snapshot of the scoring outcome on the referral, so that the basis of a recommendation remains available after weights change.	Implemented

5. Functional Requirements — Referral Lifecycle

ID	Requirement	Status
FR-LIFE-01	The system shall present each hospital with an inbox of referrals awaiting its response, showing patient, consultant, clinical detail, urgency, and time remaining before the response deadline.	Implemented
FR-LIFE-02	The system shall display the time remaining as a live countdown and shall indicate clearly when a referral is overdue.	Implemented
FR-LIFE-03	The system shall set a response deadline on every referral according to urgency: 15 minutes for emergency, 2 hours for urgent, and 24 hours for routine.	Implemented
FR-LIFE-04	The system shall provide a dedicated triage view showing only unanswered emergency referrals.	Implemented
FR-LIFE-05	The system shall require a hospital to assign a department when accepting a referral.	Implemented
FR-LIFE-06	The system shall require a hospital to give a reason when rejecting a referral, and shall make that reason visible to the consultant.	Implemented
FR-LIFE-07	The system shall check every minute for referrals whose response deadline has passed without a decision.	Implemented
FR-LIFE-08	The system shall automatically reassign an overdue referral to the next hospital in its ranked list, applying the consultant's department and doctor preference for that hospital, and setting a fresh response deadline.	Implemented
FR-LIFE-09	The system shall, for an overdue emergency referral whose ranked list is exhausted, reassign it to the nearest active hospital within 20 kilometres that has not already been tried.	Implemented
FR-LIFE-10	The system shall reject a referral automatically, with an explanatory reason, when its deadline passes and no further hospital is available.	Implemented
FR-LIFE-11	The system shall degrade the recorded performance of a hospital that allows a referral to escalate, and shall reflect that degradation in the hospital's future ranking.	Implemented
FR-LIFE-12	The system shall notify the consultant and both the previous and the new hospital when a referral escalates.	Implemented
FR-LIFE-13	The system shall maintain a clinical notes timeline on each referral, distinguishing nursing notes from consultant notes, and shall record the author and time of each note.	Implemented
FR-LIFE-14	The system shall make the clinical notes timeline visible to the referring consultant.	Implemented

ID	Requirement	Status
FR-LIFE-15	The system shall maintain the referral status through pending, accepted, rejected, admitted, and closed.	Implemented
FR-LIFE-16	The system shall allow a consultant to edit a referral while it is pending, accepted, or rejected, and shall prevent editing once the patient is admitted.	Implemented

6. Functional Requirements — Admissions and Hospital Operations

ID	Requirement	Status
FR-ADM-01	The system shall allow a hospital to admit an accepted referral, recording ward, room, bed, admitting department, and treating doctor, all of which shall be mandatory.	Implemented
FR-ADM-02	The system shall create exactly one admission record per referral.	Implemented
FR-ADM-03	The system shall allow a hospital to bill a completed case as a set of description-and-amount service lines, and shall total them automatically.	Implemented
FR-ADM-04	The system shall capture the payment method and reference on billing, and shall allow the patient bill document to be attached.	Implemented
FR-ADM-05	The system shall, on billing a case, close the referral, compute the commercial split, and create a consultant commission accrual.	Implemented
FR-ADM-06	The system shall ensure that billing a case a second time has no additional financial effect.	Implemented
FR-ADM-07	The system shall maintain a bed inventory for each hospital across ward types, recording total, occupied, and available beds.	Implemented
FR-ADM-08	The system shall allow a hospital to adjust bed counts, both incrementally and by direct entry.	Implemented
FR-ADM-09	The system shall allow a hospital to maintain the list of departments it operates, and shall use that list to determine eligibility for referrals.	Implemented
FR-ADM-10	The system shall allow a hospital to maintain a register of its doctors with specialty, registration number, consultation fee, contact details, and an availability flag.	Implemented
FR-ADM-11	The system shall provide a hospital with a financial ledger of billed cases showing the amount billed, the platform charge, and the hospital's net position, searchable and exportable.	Implemented
FR-ADM-12	The system shall allow a hospital to apply its own logo and primary colour to the portal as seen by its own users.	Implemented

7. Functional Requirements — Laboratory Module

ID	Requirement	Status
FR-LAB-01	The system shall allow a consultant to create a laboratory referral carrying patient details, an ordered list of tests, clinical notes, and attachments.	Implemented
FR-LAB-02	The system shall allocate a unique sequential code to every laboratory referral, distinct in form from hospital referral codes.	Implemented
FR-LAB-03	The system shall allow a consultant to apply a discount to a laboratory referral, capped at a maximum set for that consultant.	Implemented
FR-LAB-04	The system shall suggest laboratories for a referral and shall also allow direct selection by name.	Implemented
FR-LAB-05	The system shall require a laboratory to commit an expected report date and time when accepting a referral.	Implemented
FR-LAB-06	The system shall require a reason when a laboratory rejects a referral.	Implemented
FR-LAB-07	The system shall allow a consultant to re-refer a rejected laboratory referral to a different laboratory without re-entering patient details.	Implemented
FR-LAB-08	The system shall allow a laboratory to upload report files against a referral and shall notify the consultant on upload.	Implemented
FR-LAB-09	The system shall allow a laboratory to price each ordered test, pre-filling prices from its own catalogue where the test name matches.	Implemented
FR-LAB-10	The system shall prevent a laboratory from altering which tests were ordered by the consultant.	Implemented
FR-LAB-11	The system shall compute a laboratory bill as the gross of the priced tests less the applicable discount.	Implemented
FR-LAB-12	The system shall, on finalisation of a laboratory referral, close it, compute the commercial split, and create a consultant commission accrual.	Implemented
FR-LAB-13	The system shall maintain the laboratory referral status through pending, accepted, reported, closed, and rejected.	Implemented
FR-LAB-14	The system shall allow a laboratory to maintain a test catalogue of test name, price, and turnaround time.	Implemented

8. Functional Requirements — Commission and Settlement

ID	Requirement	Status
FR-FIN-01	The system shall compute every division of a patient bill through a single shared commission module, so that no calculation path can diverge from another.	Implemented
FR-FIN-02	The system shall support a nested commission model in which the platform takes a percentage of the bill and the consultant takes a percentage of that deduction.	Implemented
FR-FIN-03	The system shall support an additive commission model in which the consultant commission and the platform charge are computed independently of one another and summed to give the facility's liability.	Implemented
FR-FIN-04	The system shall allow the consultant commission and the platform charge each to be expressed as a percentage or as a fixed amount, independently of the other.	Implemented
FR-FIN-05	The system shall apply fixed hospital charges per referral, and fixed laboratory charges per test.	Implemented
FR-FIN-06	The system shall guarantee, under every model and combination, that the total deducted from a bill equals the consultant commission plus the platform charge.	Implemented
FR-FIN-07	The system shall never reduce the platform charge to subsidise a consultant commission.	Implemented
FR-FIN-08	The system shall resolve applicable commercial terms in the order: consultant-facility override, then facility terms, then platform defaults.	Implemented
FR-FIN-09	The system shall allow a distinct platform charge to be agreed between an individual consultant and an individual facility.	Implemented
FR-FIN-10	The system shall record on every commission accrual the model, type, rates, and amounts applied, so that historical figures remain reproducible after terms change.	Implemented
FR-FIN-11	The system shall default every existing party to the nested model, so that adoption of the additive model changes no party's terms until an administrator opts them in.	Implemented
FR-FIN-12	The system shall allow a facility to assemble a settlement from billed cases within a stated period, attach a summary document, and add notes.	Implemented
FR-FIN-13	The system shall support settlements containing consultants on different commission models, types, and rates, totalling them correctly.	Implemented

ID	Requirement	Status
FR-FIN-14	The system shall progress a settlement through pending payment, pending administrator verification, paid pending consultant payout, paid pending consultant verification, and completed.	Implemented
FR-FIN-15	The system shall allow a facility to upload a payment receipt against a settlement.	Implemented
FR-FIN-16	The system shall allow an administrator to verify a receipt, or reject it with a reason that the facility can see and act on.	Implemented
FR-FIN-17	The system shall allow an administrator to record a payout to each consultant on a settlement, with a receipt.	Implemented
FR-FIN-18	The system shall require the consultant to confirm receipt of a payout, and shall complete the settlement only when all payouts are confirmed.	Implemented
FR-FIN-19	The system shall provide a consultant with an earnings view showing wallet balance, commission by referral, and payout history, exportable as a document.	Implemented
FR-FIN-20	The system shall allow a consultant to request a withdrawal subject to a minimum amount and their available balance.	Implemented
FR-FIN-21	The system shall store every monetary value as a whole number of paisa.	Implemented
FR-FIN-22	The system shall provide the same settlement cycle for laboratories as for hospitals.	Implemented

ON THE WALLET THRESHOLD AND INITIAL HOLD

The platform settings include a wallet threshold and an initial hold, configurable by an administrator and defaulting to PKR 10,000 and PKR 9,500 respectively. These belong to an earlier automatic wallet-crediting design that has been superseded by the receipt-based settlement cycle described in FR-FIN-12 to FR-FIN-18. The settings remain configurable and the superseded module remains in the codebase but is not connected to the current billing flow. Removal is recommended in the Maintenance and Support Guide.

9. Functional Requirements — Platform Administration

ID	Requirement	Status
FR-PLAT-01	The system shall present administrators with a queue of pending registrations, including the submitted verification documents.	Implemented
FR-PLAT-02	The system shall provide registries of consultants, hospitals, and laboratories, searchable and filterable by account status.	Implemented
FR-PLAT-03	The system shall allow an administrator to correct any party's profile details.	Implemented
FR-PLAT-04	The system shall allow an administrator to configure commercial terms at platform, facility, and consultant-facility level.	Implemented
FR-PLAT-05	The system shall allow an administrator to configure the six matching-engine weights, and shall reject any configuration whose weights do not total exactly 100.	Implemented
FR-PLAT-06	The system shall allow an administrator to maintain the department catalogue and the symptom keywords that map to each department.	Implemented
FR-PLAT-07	The system shall allow an administrator to view and correct any referral, including its admission details.	Implemented
FR-PLAT-08	The system shall require multiple explicit confirmations, including entry of the patient name, before permanently deleting a referral.	Implemented
FR-PLAT-09	The system shall present bed inventory across all hospitals and allow an administrator to correct it.	Implemented
FR-PLAT-10	The system shall allow an administrator to configure platform financial defaults.	Implemented
FR-PLAT-11	The system shall allow an administrator to configure platform branding — name, logo, favicon, and colour scheme — without redeployment.	Implemented
FR-PLAT-12	The system shall record administrative actions in an audit log capturing the acting user, the action, the affected record, the network address, and the client used.	Implemented
FR-PLAT-13	The system shall allow the audit log to be searched and exported.	Implemented
FR-PLAT-14	The system shall provide administrators with platform-wide analytics covering referral volume, activity, and bed availability.	Implemented
FR-PLAT-15	The system shall allow an administrator to manage the administrator team.	Implemented

10. Functional Requirements — Notifications, Documents, and Payments

ID	Requirement	Status
FR-NOTE-01	The system shall raise notifications for registration, referral, admission, billing, settlement, payout, and account events.	Implemented
FR-NOTE-02	The system shall deliver notifications in the application, with an unread counter, and shall deliver them without the user refreshing the page.	Implemented
FR-NOTE-03	The system shall deliver notifications by email.	Implemented
FR-NOTE-04	The system shall deliver notifications by WhatsApp where a mobile number is available.	Partial — implemented; requires client completion of message template approval
FR-NOTE-05	The system shall deliver a facility notification to the facility owner and to every active team member of that facility.	Implemented
FR-NOTE-06	The system shall exclude personal data from stored notification payloads.	Implemented
FR-NOTE-07	The system shall ensure that failure of one notification channel does not prevent delivery on the others.	Implemented
FR-NOTE-08	The system shall update inboxes, dashboards, and bed counts in place as underlying events occur.	Implemented
FR-DOC-01	The system shall generate downloadable records for referrals, consultant and hospital rosters, laboratory records, and party profiles.	Implemented
FR-DOC-02	The system shall include uploaded attachments within generated referral documents, embedding images and merging attached documents.	Implemented
FR-DOC-03	The system shall allow a consultant to export an earnings statement.	Implemented
FR-DOC-04	The system shall allow a hospital to export its financial ledger in a spreadsheet-compatible format.	Implemented
FR-DOC-05	The system shall accept uploaded files of image and document type only, subject to a maximum size, and shall store them in cloud storage.	Implemented

ID	Requirement	Status
FR-DOC-06	The system shall support online payment through a payment gateway, with cryptographic signing of requests and verification of callbacks.	Partial — implemented; requires client live merchant credentials

11. Non-Functional Requirements

11.1 Security

ID	Requirement	Status
NFR-SEC-01	Passwords shall be stored only as salted cryptographic hashes, never in recoverable form.	Implemented
NFR-SEC-02	Patient national identity numbers shall be encrypted before storage.	Implemented
NFR-SEC-03	All API access shall require a valid authentication credential, except for registration, sign-in, password reset, public branding, and the payment gateway callback.	Partial — the file upload endpoint is currently reachable without authentication and is scheduled for correction
NFR-SEC-04	Every endpoint shall enforce the role permitted to call it.	Implemented
NFR-SEC-05	Authentication endpoints shall be rate limited to resist automated guessing.	Implemented
NFR-SEC-06	All other endpoints shall be rate limited.	Not implemented — recommended
NFR-SEC-07	Standard protective HTTP response headers shall be applied to all responses.	Implemented
NFR-SEC-08	Cross-origin access shall be restricted to configured origins in production.	Partial — supported, but permissive if left unconfigured; must be set at deployment
NFR-SEC-09	Payment gateway callbacks shall be verified cryptographically before being acted upon.	Implemented
NFR-SEC-10	Administrative actions shall be recorded in an immutable-by-convention audit log.	Implemented
NFR-SEC-11	Password reset and email verification credentials shall be single use and time limited.	Implemented

ID	Requirement	Status
NFR-SEC-12	Patient identity encryption shall provide tamper detection as well as confidentiality.	Not implemented — the mode in use provides confidentiality only; upgrade recommended

11.2 Performance and Scalability

ID	Requirement	Status
NFR-PERF-01	Hospital ranking shall be computed on demand and returned within an interactive response time for a metropolitan-scale hospital set.	Implemented
NFR-PERF-02	The escalation check shall run every minute and shall complete within its interval.	Implemented
NFR-PERF-03	Frequently queried fields shall be indexed, including geographic coordinates, user role and status, and notification recipient.	Implemented
NFR-PERF-04	Client-side caching shall prevent redundant repeat requests for unchanged data.	Implemented
NFR-PERF-05	Real-time events shall be addressed to specific users, roles, or facilities rather than broadcast to all connections.	Implemented
NFR-PERF-06	The backend shall be capable of running as more than one concurrent instance.	Not implemented — one-time-password state is held in process memory, which constrains the system to a single instance

11.3 Reliability and Integrity

ID	Requirement	Status
NFR-REL-01	Billing a case shall be idempotent, so that a repeated request cannot duplicate a commission accrual.	Implemented
NFR-REL-02	Referral code allocation shall be atomic, so that concurrent referrals cannot receive the same code.	Implemented
NFR-REL-03	Failure of a notification channel shall not fail the operation that raised the notification.	Implemented

ID	Requirement	Status
NFR-REL-04	A failed backend process shall restart automatically.	Implemented
NFR-REL-05	Monetary arithmetic shall not be subject to floating-point rounding error.	Implemented
NFR-REL-06	Commercial terms in force at the time of a calculation shall be recoverable afterwards.	Implemented

11.4 Usability and Accessibility

ID	Requirement	Status
NFR-USE-01	The interface shall be usable on desktop, tablet, and mobile without a separate application.	Implemented
NFR-USE-02	Complex data entry shall be presented as guided steps with visible progress.	Implemented
NFR-USE-03	The interface shall offer a light and a dark appearance, remembered per browser.	Implemented
NFR-USE-04	Every action shall produce visible confirmation of success or failure.	Implemented
NFR-USE-05	Time-critical information shall be presented as a live countdown rather than a static timestamp.	Implemented
NFR-USE-06	The platform operator and individual hospitals shall be able to apply their own branding without redeployment.	Implemented

11.5 Maintainability

ID	Requirement	Status
NFR-MNT-01	The backend shall separate routing, request handling, business logic, and data access into distinct layers.	Implemented
NFR-MNT-02	Financially material logic shall be covered by automated tests.	Implemented
NFR-MNT-03	All configuration and secrets shall be supplied by environment settings, never committed to source.	Implemented

ID	Requirement	Status
NFR-MNT-04	Every configuration setting the system reads shall be documented.	Partial — the payment gateway and messaging settings are documented in a separate runbook rather than the main configuration template; consolidation recommended
NFR-MNT-05	Controllers, routes, and scheduled jobs shall be covered by integration tests.	Not implemented — recommended

12. Requirements Summary

Area	Implemented	Partial	Not implemented	Total
Identity and access	23	0	0	23
Referral creation and matching	19	0	0	19
Referral lifecycle	16	0	0	16
Admissions and hospital operations	12	0	0	12
Laboratory module	14	0	0	14
Commission and settlement	22	0	0	22
Platform administration	15	0	0	15
Notifications, documents, payments	12	2	0	14
Non-functional	28	4	4	36
Delivered scope total	161	6	4	171

Within the delivered scope, 161 of 171 requirements are fully implemented. Six are partial: two await a client action to activate an optional integration, and four are non-functional items requiring configuration or a documented improvement. Four are not implemented, all non-functional, and each is recorded with a recommendation in the Security and Compliance document.